

SNEHA SURESH ZALTE

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PROFESSIONAL SUMMARY

Results-oriented MCA candidate with hands-on experience in Machine Learning, MLOps, and cloud-based deployments. Skilled in building end-to-end ML pipelines, developing REST APIs, and deploying AI-driven applications on AWS and GCP. Demonstrates expertise in real-time communication systems and accessibility solutions, with a proven ability to deliver production-ready solutions across full-stack and AI domains.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, XML, JavaScript
Frameworks & Libraries: Flask, MediaPipe, OpenCV, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Keras, Hugging Face, React
AI / ML: Machine Learning, Deep Learning, NLP, Computer Vision, Speech-to-Text (STT), Text-to-Speech (TTS), Model Training & Fine-tuning, Feature Engineering, Data Preprocessing
MLOps: MLflow, Model Versioning, Experiment Tracking, CI/CD Pipelines, Pipeline Orchestration, Model Deployment & Monitoring
Cloud Platforms: AWS (EC2, S3, Lambda, SageMaker, ECS), GCP (Vertex AI, GCS, Compute Engine, Cloud SQL), Azure (Fundamentals)
DevOps & Containers: Docker, Docker Compose, Kubernetes (Basics), Git, GitHub Actions
Databases: MySQL, MongoDB, Firebase Realtime DB, PostgreSQL
Tools & Platforms: Android Studio, Postman, Linux, Microsoft Purview, Microsoft Sentinel

WORK EXPERIENCE

Junior Software Engineer | Ant Systemz Pvt Ltd Oct 2026 – Present

- Built and fine-tuned production-grade Speech-to-Text (STT) and Text-to-Speech (TTS) models, improving inference reliability for real-world use cases.
- Designed and implemented end-to-end ML pipelines covering data preprocessing, feature engineering, model training, evaluation, and deployment.
- Applied MLOps best practices including model versioning with MLflow, experiment tracking, CI/CD pipelines via GitHub Actions, and production model monitoring.
- Developed and deployed REST APIs for model serving and real-time inference, integrated with frontend applications built in React and JavaScript.
- Containerized ML workloads using Docker and Docker Compose; enabled scalable inference service deployment.
- Leveraged GCP services (Vertex AI, GCS, Compute Engine, Cloud SQL/PostgreSQL) for the full ML lifecycle from training to deployment.
- Utilized AWS (Lambda, ECS) for lightweight container orchestration and serverless triggers.
- Implemented endpoint data security using Microsoft Purview (Endpoint DLP) and Microsoft Sentinel for threat detection, monitoring, and compliance.

EDUCATION

Master of Computer Applications (MCA) Navinchandra Mehta Institute of Technology & Development, Mumbai	2024 – 2026 (Expected)
Bachelor of Computer Applications (BCA) SNDT Women's University, Mumbai CGPA: 9.08 / 10	2021 – 2024
Higher Secondary Certificate (HSC – Science) B.K. Birla College, Kalyan 92%	2019 – 2021
Secondary School Certificate (SSC) I.E.S. School, Titwala 89.04%	2019

PROJECTS

SignConnect – Bridging Communication Through Sign Language

- Developed a real-time video communication platform enabling seamless interaction between deaf and hearing users.
- Integrated MediaPipe for hand gesture detection and AI-driven sign-to-text and voice conversion pipelines.
- Architected with Flask, WebRTC, and Socket.IO for low-latency, cross-device real-time communication.
- Achieved over 60% gesture recognition accuracy during testing, demonstrating model effectiveness in real-world scenarios.

Tech Stack: Python, Flask, JavaScript, WebRTC, Socket.IO, MediaPipe, OpenCV, gTTS

Learn-In – An Android-Based E-Learning Platform

- Designed and developed an Android e-learning application with an intuitive UI targeting students and educators.
- Integrated Firebase Realtime DB for live data synchronization and user authentication.
- Delivered an accessible, efficient learning environment validated by 50+ beta testers.

Tech Stack: Android Studio, XML, Java, Firebase

CERTIFICATIONS

- Data Science Certification
- Course on Computer Concepts (CCC)

ACHIEVEMENTS & LEADERSHIP

- Awarded by SNDT Women's University for developing an Inspiring Project (2024).
- Served as Head of Student Council, demonstrating leadership, communication, and organizational skills.

LANGUAGES

English (Fluent) | Hindi (Fluent) | Marathi (Proficient)